

Markscheme

Mathematics: analysis and approaches
Practice question bank

172. (a)

Consider the system $2x + y - z = 3$, $x - y + 2z = 8$, $3x + 2y - z = 7$.

Solve the system for x , y and z .

subtract equation (1) from equation (3): $x + y = 4$ (M1) A1

from (1): $z = 2x + y - 3$; substitute into (2): $x - (4 - x) + 2(2x + y - 3) = 8$ (M1)

solving simultaneously gives (M1)

$x = 2.5$, $y = 1.5$, $z = 3.5$ A1

[5 marks]